

DSA Day 28: Heaps, Heap Sort & Priority Queue

Book: Data Structures Through C in Depth - Ch6 §6.16 · pp 277-286 | Code in Python

Overview

A binary heap is a complete tree in an array with the heap property. sift-up on insert, sift-down on delete; heap sort sorts in place.

Key concepts to master

- Heap property
- sift-up / sift-down
- buildHeap $O(n)$
- heapSort
- priority queue

Python implementation

```
import heapq
a=[4,10,3,5,1]; heapq.heapify(a)
out=[heapq.heappop(a) for _ in range(len(a))] # sorted ascending
```

Complexity

insert/delete $O(\log n)$; buildHeap $O(n)$; heapSort $O(n \log n)$.

Common pitfalls

- Mixing max-heap vs min-heap
- Breaking the complete-tree shape

Practice

- Build a heap from {4,10,3,5,1} and heap-sort
- Implement a priority queue

